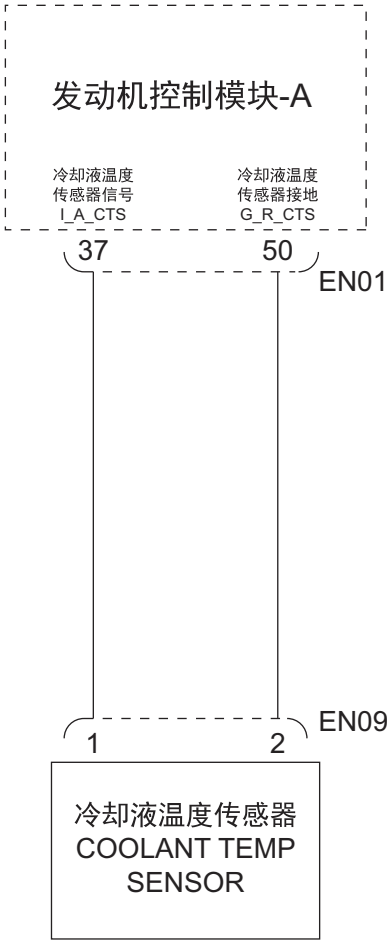
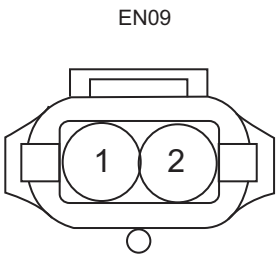
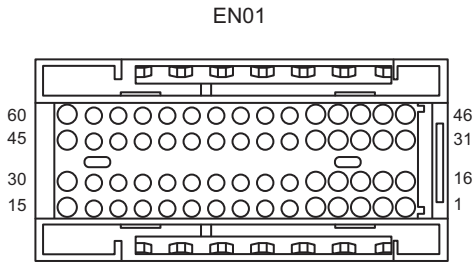
P0116、P0117、P0118

Description of fault diagnosis codes
P0116-27：defect fault check for Coolant temperature dynamic plausibility test
P0117-11：SRC low for Engine coolant temperature(down stream)
P0118-12：SRC High for Engine coolant temperature(down stream)
Detection tools
1 Multimeter and diagnostic apparatus
Possible causes
• Fault in harness or harness plug • Coolant temperature sensor fault
Description of circuit diagram
The engine coolant temperature sensor is a variable resistor used to measure the engine coolant temperature. The engine control module provides 5V voltage to the signal circuit of engine coolant temperature sensor. It provides bonding to low level reference circuit. This diagnosis will inspect the open circuit between the engine control module and engine coolant temperature sensor, short circuit to bonding or intermittent circuit fault.

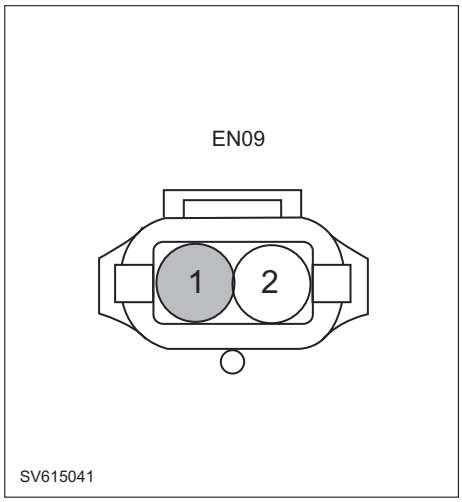
Circuit diagram

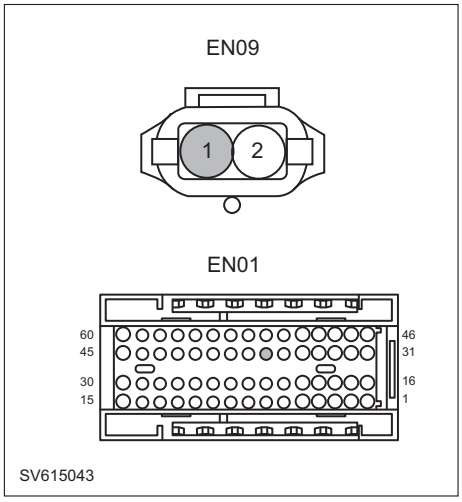
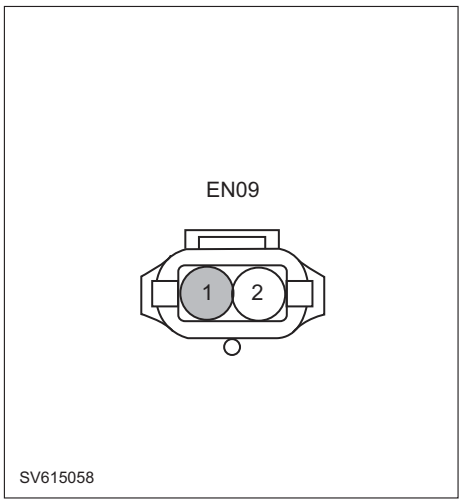


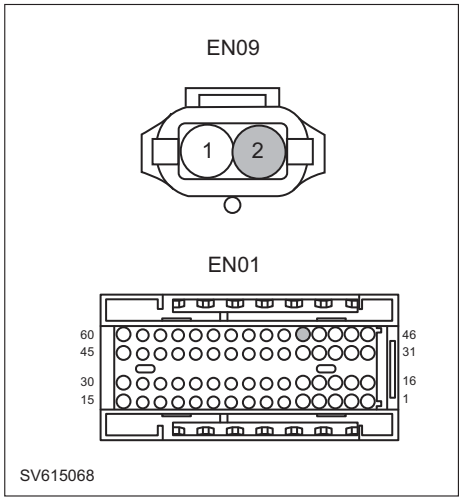
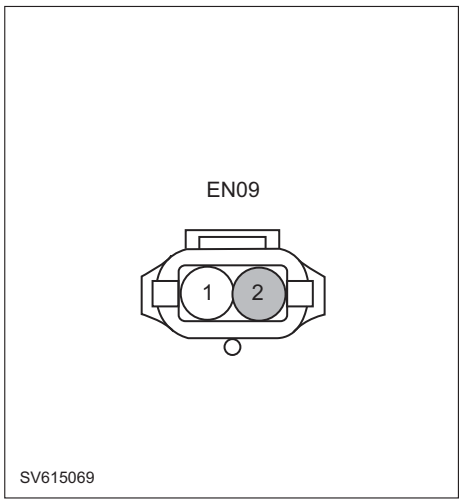




SV615026

Troubleshooting procedures		
Step 1	General inspection.	
		1 Check the fuse for damage. 2 Check battery capacity. 3 Check the harness plugs of engine control module and oil metering unit for damage, poor contact, aging and looseness. 4 Check whether the result is normal.
		No → Repair or replace the faulted parts.
No ↓		
Step 2	Check fault codes.	
		1 Connect the fault diagnosis instrument to diagnosis interface. 2 Turn the ignition switch to ON position. 3 Diagnose the engine control system with diagnostic apparatus. 4 Is there any other fault code except for P0116,P0117,P0118?
		Yes → Reference: DTC Summary List (EU5).
No ↓		
Step 3	Check the signal circuit of coolant temperature sensor.	
		1 Turn the ignition switch to OFF. 2 Disconnect the harness plug of coolant temperature sensor. 3 Turn the ignition switch to ON position. 4 Measure the voltage between the harness plug of coolant temperature sensor and ground. Measuring circuit: the voltage between EN09 harness plug terminal 1 and ground. **Standard value:4.8~5.2V** 5 Check whether the result is normal?
		Yes → Go to Step 6.
No ↓		

Step 4	Check the signal circuit of coolant temperature sensor(open circuit).
	1 Turn the ignition switch to OFF. 2 Disconnect the harness plug of engine control module. 3 Measure the resistance of harness plug between coolant temperature sensor and engine control module. Measuring circuit: the resistance between EN09 plug terminal 1 and EN01 plug terminal 37. **Standard value: ≈ 0Ω** 4 Check whether the result is normal?
	No → Repair or replace the faulted parts.
Yes ↓	
Step 5	Check the signal circuit of coolant temperature sensor(short circuit).
	1 Measure the resistance between the harness plug of coolant temperature sensor and ground. Measure circuit: the resistance between EN09 plug terminal 1 and ground. **Standard value: ∞** 2 Measure the voltage between the harness plug of coolant temperature sensor and ground. Measure circuit: the voltage between EN09 plug terminal 1 and ground. **Standard value: 0V** 3 Check whether the result is normal?
	No → Repair or replace the faulted parts.
Yes ↓	

Step 6	Check the ground circuit of coolant temperature sensor(open circuit).
	1 Turn the ignition switch to OFF. 2 Disconnect the harness plug of engine control module. 3 Measure the resistance of harness plug between coolant temperature sensor and engine control module. Measuring circuit: the resistance between EN09 plug terminal 2 and EN01 plug terminal 50. **Standard value: ≈ 0Ω** 4 Check whether the result is normal?
	No → Repair or replace the faulted parts.
Yes ↓	
Step 7	Check the ground circuit of coolant temperature sensor(short circuit).
	1 Measure the resistance between the harness plug of coolant temperature sensor and ground. Measure circuit: the resistance between EN09 plug terminal 2 and ground. **Standard value: ∞** 2 Measure the voltage between the harness plug of coolant temperature sensor and ground. Measure circuit: the voltage between EN09 plug terminal 2 and ground. **Standard value: 0V** 3 Check whether the result is normal?
	No → Repair or replace the faulted parts.
Yes ↓	
Step 8	Check the coolant temperature sensor (components).
	1 Measure the resistance value of coolant temperature sensor terminal. Measure circuit: the resistance between terminal 1 and terminal 2 of coolant temperature sensor. **Standard value: 2.3~2.6KΩ (20 ℃)** **Standard value: 1.1~1.2KΩ (40 ℃)** **Standard value: 0.5~0.65KΩ (60 ℃)** 2 Check whether the result is normal?
	No → Replace the coolant temperature sensor.

Next ↓	
Step 7	Complete the check and confirm whether the system works normally nor not.